No.	Status	Date
HIS-2007-0388-DS-3-RV-101	Issued	2007-09-18

UT Thickness Data Sheet - 3-RV-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 3-RV-101 located within the ATU-100. The scope is limited to the thickness data for this specific line as part of the broader survey documented in HIS-2007-0388.

Kevin Doucette, acting as the lead UT technician, is responsible for the execution of the measurements and the recording of the data. All activities were performed in accordance with the technical requirements of the project.

Technical Basis and Procedures

Thickness measurements were performed under the site NDE procedure QSCP-NDT-001 Rev 0 and the company procedure HIS-UT-002 Rev 1. These documents govern the methodology for ultrasonic thickness measurement on the site.

The inspection techniques and equipment calibration were conducted in accordance with ASTM E797.

Line Specifications

Line 3-RV-101 is a regenerator overhead line constructed from A106 Gr.B. The line has an outer diameter of 88.9 mm and a nominal thickness of 7.62 mm.

The design limits for this line are 345 kPa(g) and 121 °C. The minimum allowable thickness (t-min) is 3.56 mm.

Assessment Criteria

The retirement thickness for 3-RV-101 is defined as the t-min of 3.56 mm. Any reading recorded below this value is subject to an NCR.

Corrosion rates are determined by the larger of the long-term rate or the short-term rate over a period of at least 3 years. Remaining life is calculated as (t - t-min) / rate.

Inspection Interval

The next thickness reading is scheduled for no more than half the remaining life, subject to the class maximum. 3-RV-101 is designated as Class 1 due to its service as regenerator overhead sour gas.

The class maximum for thickness and external visual inspections for Class 1 is 5 years / 5 years.

Thickness readings, measured 2007-09-18

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2006-02-24	6.98	6.95	0.056	0.047	>50	2012-09-18	
3-RV-101-02	7.62	3.56	2006-02-24	6.93	6.89	0.059	0.064	>50	2012-09-18	
3-RV-101-03	7.62	3.56	2006-02-24	6.57	6.56	0.083	0.090	33.3	2012-09-18	
3-RV-101-04	7.62	3.56	2006-02-24	6.65	6.66	0.081	0.079	38.4	2012-09-18	
3-RV-101-05	7.62	3.56	2006-02-24	7.01	6.87	0.069	0.058	47.7	2012-09-18	
3-RV-101-06	7.62	3.56	2006-02-24	6.74	6.71	0.076	0.075	41.3	2012-09-18	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2006-02-24	4.85	4.83	0.061	0.070	29.0	2012-09-18	
2-RV-102-02	5.54	2.79	2006-02-24	4.84	4.67	0.080	0.069	23.6	2012-09-18	
2-RV-102-03	5.54	2.79	2006-02-24	5.14	5.01	0.041	0.035	>50	2012-09-18	
2-RV-102-04	5.54	2.79	2006-02-24	5.07	5.12	0.040	0.032	>50	2012-09-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2007-09-18	_____